The Impact of Artificial Intelligence on Primary School Students' Motivation and Engagement: A Systematic Review

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Abstract

The application of artificial intelligence (AI) in primary education has attracted growing attention, with increasing studies exploring its potential to enhance learning. However, prior reviews often overlook primary school students and conceptualize motivation and engagement as single-dimensional. This systematic review, guided by Self-Determination Theory and the multidimensional engagement framework, analyzed 33 empirical studies to evaluate AI's impact on primary students' motivation and engagement. Findings show that AI can fulfil students' basic psychological needs for autonomy, competence, and relatedness, and effectively promote general, intrinsic, and extrinsic motivation (particularly external and identified regulation) among primary school students. AI also enhances engagement across behavioral, cognitive, emotional, and agentic dimensions. However, when AI responses are complex, mechanical, or less engaging than human-mediated tasks, it may lead to negative experiences including frustration, boredom, or reduced motivation. AI's impact varies across contexts by satisfying the need for competence and fostering intrinsic motivation and emotional engagement in STEM subjects, while promoting relatedness in humanities and social sciences. Younger students generally show greater motivation and engagement with AI-supported learning than older ones. Different types of AI tools'personalized, emotionally supportive, interactive, immersive, and feedback-oriented AI'offer distinct advantages in fostering motivation and engagement. Compared to traditional AI, generative AI uniquely enhances learning value perception and strategy adjustment. This review provides theoretical and empirical support for integrating AI in primary education and highlights the need to prioritize motivation and engagement in AI-supported instructional design.